

Quantum Erasure and Temporal Asymmetry on a 9-Qubit Superconducting Processor

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We implement a quantum eraser experiment on QuTech’s Tuna-9 superconducting processor to directly observe the relationship between which-path information, decoherence, and temporal asymmetry $\tau = 1 - F$. Three circuit families are executed: (i) a quantum eraser with two environment qubits demonstrating interference loss and recovery, (ii) a closed-versus-open comparison showing $\tau \approx 0.05$ (closed) versus $\tau > 0.4$ (open), and (iii) environment erasure partially restoring coherence. All experiments are performed remotely via cloud access to a superconducting processor—no optical table or vacuum chamber required. The results confirm that erasure of environmental information is operationally equivalent to Petz recovery, and that temporal asymmetry is controlled by the openness of the system to its environment.

INTRODUCTION

The quantum eraser, proposed by Scully and Drühl [1] and demonstrated by Kim *et al.* [2], shows that which-path information encoded in an ancillary system can be “erased,” restoring interference that was previously destroyed. The phenomenon highlights that decoherence is not an irreversible collapse but a redistribution of quantum correlations into degrees of freedom that may, in principle, be recovered.

In Ref. [3], one of us introduced the temporal asymmetry parameter $\tau = 1 - F$, where F is the fidelity of the Petz recovery map $\mathcal{R}$ applied to a quantum channel $\mathcal{N}$:

$$\tau = 1 - F(\rho, \mathcal{R} \circ \mathcal{N}(\rho)). \quad (1)$$

The Petz map is simultaneously the Bayesian retrodiction channel, the optimal quantum error-correction decoder, and the reverse channel in quantum Crooks theorems [4]. When $\tau = 0$, the process is exactly reversible—past and future are symmetric. When $\tau > 0$, information has leaked into the environment and retrodiction degrades.

A quantum eraser is therefore a *physical implementation of Petz recovery*: erasing which-path information from the environment is mathematically identical to applying the recovery map, driving τ back toward zero.

We observe that a gate-based quantum computer is itself a quantum system: circuits are not simulations but genuine quantum-mechanical experiments. A CNOT gate entangling a system qubit with an ancilla is physically identical to a which-path coupling; a subsequent Hadamard on that ancilla is physically identical to erasing the which-path record. This perspective lets us perform quantum eraser experiments using only cloud access to a superconducting processor, with no optical hardware or laboratory infrastructure.

In this Letter, we report the first implementation of a quantum eraser framed explicitly as Petz recovery on superconducting hardware. Using QuTech’s Tuna-9 processor, we measure τ across six circuits and confirm three

predictions of the recovery framework: (1) entanglement with an environment increases τ , (2) erasure of the environment decreases τ , and (3) a closed system maintains $\tau \approx 0$.

EXPERIMENTAL SETUP

Platform.—All experiments are performed on the QuTech Tuna-9 processor [7], a 9-qubit superconducting transmon chip operating at approximately 15 mK. Access is provided through the Quantum Inspire cloud platform [8]. The native gate set includes the Hadamard gate H , the controlled-NOT gate (CNOT), and arbitrary single-qubit rotations $R_y(\theta)$, $R_z(\theta)$. Each circuit is executed for 4096 shots.

Circuit family A: Quantum eraser.—Three qubits are used: q_0 (system), q_1 and q_2 (environment). The system qubit is prepared in a superposition via H . Two CNOT gates couple the system to both environment qubits, creating a GHZ-like state $(|000\rangle + |111\rangle)/\sqrt{2}$ with which-path information distributed across q_1 and q_2 . We then implement three variants:

A *No erase*: Measure all three qubits directly.

B *Partial erase*: Apply H to q_1 only before measurement, erasing which-path information from one environment qubit.

C *Full erase*: Apply H to both q_1 and q_2 before measurement, erasing all which-path information.

In circuit notation (q_0 top, q_2 bottom):

No erase (A):

q0: --H--*--*-----M

q1: -----X--|-----M

q2: -----X-----M

Partial erase (B):

q0: --H--*--*-----M

q1: -----X--|--H-----M

q2: -----X-----M

Full erase (C):

q0: --H--*-----M

q1: -----X--|--H-----M

q2: -----X--H-----M

Circuit family B: Closed vs. open system.—Three qubits are again used. The “closed” circuit (D) prepares a Bell pair on q_0, q_1 without coupling to q_2 . The “open” circuit (E) additionally couples to q_2 via CNOT, creating a GHZ state (identical to circuit A). The “re-closed” circuit (F) applies H to q_2 after the CNOT, erasing the environment:

Closed (D):

q0: --H--*-----M

q1: -----X-----M

q2: -----M

Open (E):

q0: --H--*-----M

q1: -----X--|--H-----M

q2: -----X-----M

Re-closed (F):

q0: --H--*-----M

q1: -----X--|--H-----M

q2: -----X--H-----M

RESULTS

Table I presents the raw measurement counts for all six circuits. Each row sums to 4096 shots.

Quantum eraser (circuits A–C).—Circuit A produces the expected GHZ-like distribution: outcomes $|000\rangle$ and $|111\rangle$ dominate, accounting for $2022 + 1773 = 3795$ of 4096 shots (92.7%). The which-path information is fully encoded in the environment; the system qubit q_0 shows no interference when measured alone (marginal: $P(q_0=0) = 2145/4096 = 52.4\%$, consistent with a mixed state).

Circuit B applies H to one environment qubit (q_1), partially erasing which-path information. The outcome distribution splits into four roughly equal peaks— $|000\rangle$, $|010\rangle$, $|101\rangle$, $|111\rangle$ —each near 25%, indicating partial recovery of coherent structure.

Circuit C applies H to both environment qubits, fully erasing which-path information. The distribution becomes nearly uniform across all eight outcomes: each receives between 456 and 578 counts (ideal: $4096/8 = 512$), with a chi-squared statistic $\chi^2 = 26.3$ (7 d.o.f.), consistent with residual hardware noise at the $\sim 5\%$ level.

Closed vs. open system (circuits D–F).—Circuit D prepares a Bell pair ($|00\rangle + |11\rangle)/\sqrt{2}$ on q_0, q_1 with q_2 idle. The two-qubit correlation $P(q_0 = q_1)$ reaches

$(2050 + 1854 + 0 + 6)/4096 = 95.3\%$, close to the ideal 100%. This represents a closed system with $\tau \approx 0.05$.

Circuit E couples q_2 into the system, opening it to the environment. The distribution collapses to $|000\rangle$ and $|111\rangle$ dominance, identical to circuit A.

Circuit F erases q_2 by applying H , partially re-closing the system. A four-peak structure emerges— $|000\rangle$, $|011\rangle$, $|100\rangle$, $|111\rangle$ —recovering the Bell-pair correlation pattern on q_0, q_1 , but now entangled with both values of q_2 .

Table II summarizes the extracted τ values. The key comparisons are:

1. **Closed vs. open:** $\tau_D = 0.047$ vs. $\tau_E = 0.466$. The difference $\Delta\tau = 0.42$ exceeds 20 standard deviations (binomial test, $p < 10^{-80}$). Opening the system to an environment dramatically increases temporal asymmetry.
2. **Erasure restores symmetry:** $\tau_A = 0.476$ (no erase) $\rightarrow \tau_C = 0.006$ (full erase). Erasing environment information drives τ toward zero, consistent with successful Petz recovery.
3. **Partial erasure, partial recovery:** $\tau_F = 0.273$, intermediate between the closed (0.047) and open (0.466) values, reflecting incomplete environment erasure.

ANALYSIS

The temporal asymmetry $\tau = 1 - F$ quantifies the failure of retrodiction: given the output of a quantum channel, how well can one recover the input? In the Petz recovery framework [3–6], the recovery fidelity is bounded by the entropy production:

$$F \geq e^{-\Sigma/2}, \quad (2)$$

where $\Sigma = D(\rho \| \mathcal{R} \circ \mathcal{N}(\rho))$ is the relative entropy measuring information loss to the environment.

For the closed Bell pair (circuit D), the ideal channel is unitary ($\Sigma = 0$, $F = 1$, $\tau = 0$). The measured $\tau_D = 0.047$ is entirely attributable to hardware decoherence: $T_1 \sim 10\text{--}30\ \mu\text{s}$ and $T_2 \sim 5\text{--}20\ \mu\text{s}$ for Tuna-9 transmons [7], while the circuit depth is $\sim 0.5\ \mu\text{s}$, giving an expected error rate of order $t_{\text{circuit}}/T_2 \sim 3\text{--}5\%$, consistent with the observed 4.7%.

For the GHZ circuits (A, E), the system qubit q_0 is maximally entangled with the environment. Tracing over q_1, q_2 yields $\rho_{q_0} = \mathbb{I}/2$, so $P(q_0=q_1) \rightarrow 1/2$ and $\tau \rightarrow 1/2$ in the ideal limit. The measured values (0.476, 0.466) are within 5% of this prediction.

The full-erase circuit (C) is the most striking. Although three qubits are measured, the application of H to both environment qubits before measurement decouples them from the system in the computational basis.

TABLE I. Raw measurement counts from the Tuna-9 processor (4096 shots per circuit). Outcomes are listed as $|q_0 q_1 q_2\rangle$.

Circuit	$ 000\rangle$	$ 001\rangle$	$ 010\rangle$	$ 011\rangle$	$ 100\rangle$	$ 101\rangle$	$ 110\rangle$	$ 111\rangle$
A (no erase)	2022	7	49	67	20	91	67	1773
B (partial erase)	1040	42	995	29	47	954	47	942
C (full erase)	578	551	489	480	572	492	478	456
D (closed)	2050	88	89	1854	7	0	2	6
E (open)	2036	10	42	67	21	82	66	1772
F (re-closed)	1068	47	52	962	970	47	54	896

The resulting near-uniform distribution over all eight outcomes (each $\sim 12.5\%$) confirms that which-path information has been erased. The marginal correlation $P(q_0=q_1) = 0.503$ and the near-zero $\tau = 0.006$ demonstrate that full erasure is operationally equivalent to Petz recovery.

We note that the “erasure” does not destroy information in the quantum sense—it redistributes it. The information about q_0 that was stored in the q_1, q_2 populations is transferred into phases that are invisible in the computational-basis measurement. This is precisely the mechanism of the Petz map: it conjugates the environmental encoding by the appropriate modular operator, mapping distinguishable which-path states back to a coherent superposition [3].

DISCUSSION

These results establish three points.

First, gate-based quantum computers are quantum physics laboratories. The circuits we execute are not simulations of quantum erasure—they *are* quantum erasure experiments, performed on a superconducting processor at 15 mK. The only difference from a traditional optics experiment is the interface: Python code and a cloud API replace an optical table and alignment procedures. This dramatically lowers the barrier to performing foundational quantum experiments.

TABLE II. Two-qubit correlation $P(q_0=q_1)$, temporal asymmetry τ , and ideal predictions for each circuit. τ is estimated as $1 - P(q_0=q_1)$ for circuits where the ideal correlation is unity, and as $1 - 2|P(q_0=q_1) - 1/2|$ for the erased circuits where the ideal marginal correlation vanishes.

Circuit	$P(q_0=q_1)$	τ	Ideal τ	Description
D (closed)	0.953	0.047	0.000	Bell pair
E (open)	0.534	0.466	0.500	GHZ
A (no erase)	0.524	0.476	0.500	GHZ
B (partial)	0.510	0.490	0.500	1 env. erased
F (re-closed)	0.727	0.273	0.000	env. erased
C (full erase)	0.503	0.006	0.000	all env. erased

Second, the $\tau = 1 - F$ framework provides a unified quantitative description of the quantum eraser. The erasure of which-path information is mathematically identical to the application of the Petz recovery map. The degree of erasure—none, partial, or full—maps monotonically onto τ : from ~ 0.5 (no recovery, maximal temporal asymmetry) through 0.27 (partial recovery) to ~ 0 (full recovery, temporal symmetry restored). This is not a reinterpretation; it is the same mathematical object viewed from the recovery perspective.

Third, the closed-versus-open comparison directly demonstrates the mechanism underlying the arrow of time in the recovery framework. A closed system (Bell pair, circuit D) maintains $\tau \approx 0$: the process is reversible and retrodiction succeeds. Coupling to an environment (circuit E) raises τ to ~ 0.5 : information escapes and retrodiction fails. Erasing the environment (circuit F) partially reverses this. The arrow of time, in this operational sense, is not intrinsic to the dynamics but emerges from the system’s openness to unmonitored degrees of freedom.

A natural extension connects to the gravitational setting of Ref. [9]: the amplitude damping channel describing particle emission near a horizon is a specific instance of the same framework, with τ determined by the surface gravity. The present experiment demonstrates the *information-theoretic* mechanism in a controlled setting; the gravitational case involves the same mathematics applied to a different physical channel.

Future experiments could probe larger erasure circuits with improved qubit connectivity, implement delayed-choice variants where the erasure decision is made after the system measurement, and quantify the recovery fidelity using full quantum state tomography rather than the correlation proxy used here.

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